

30-7-23

Physics

Ans. 1 Convex lens is used to produce real image. Ans.

Ans. 2 Power (P) = $\frac{1}{\text{Focal length (f) in m}}$

In Case I ÷

'f = 10 cm given

$$f = \frac{10}{100} = 0.1 \text{ m}$$

$$P = \frac{1}{0.1} = 10 \text{ D}$$

In Case II

F = 20 cm

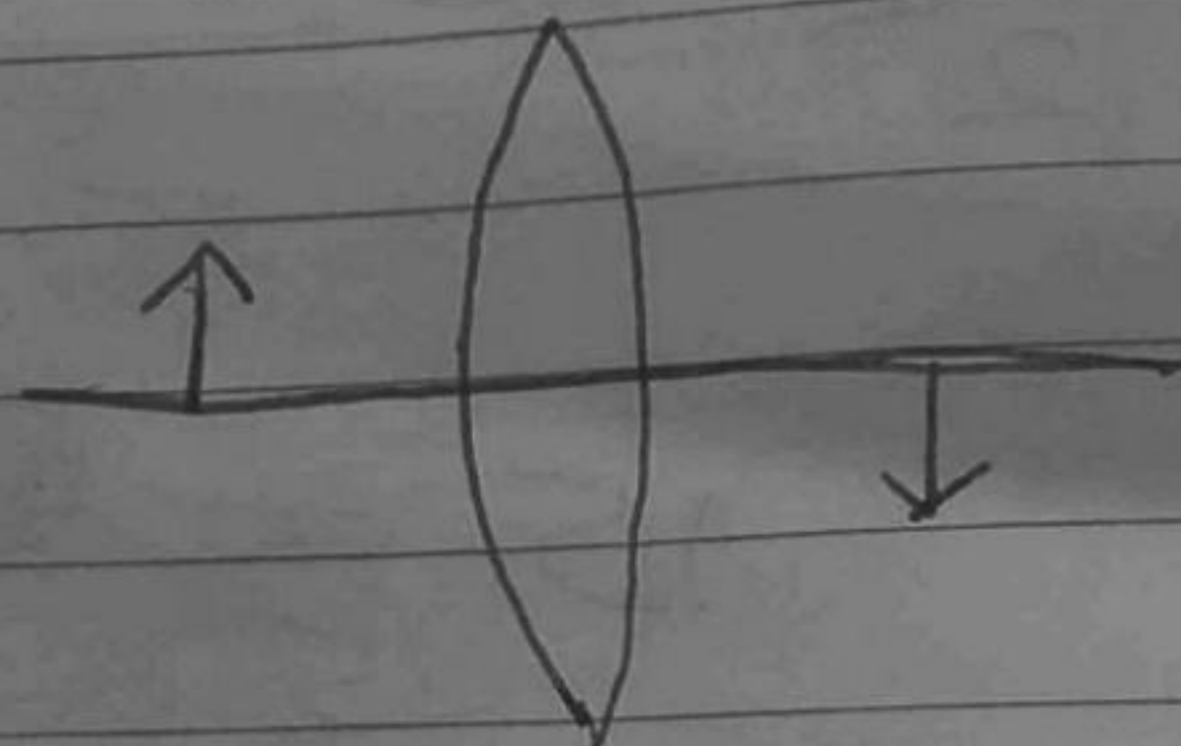
$$F = \frac{20}{100} = 0.2$$

$$P = \frac{1}{0.2} = 5 \text{ D}$$

Ans. 3

$$U = -5 \text{ cm}$$

$$V = +10 \text{ cm}$$



$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} + \frac{1}{5}$$

$$\frac{1}{f} = \frac{1+2}{10} = \frac{3}{10}$$

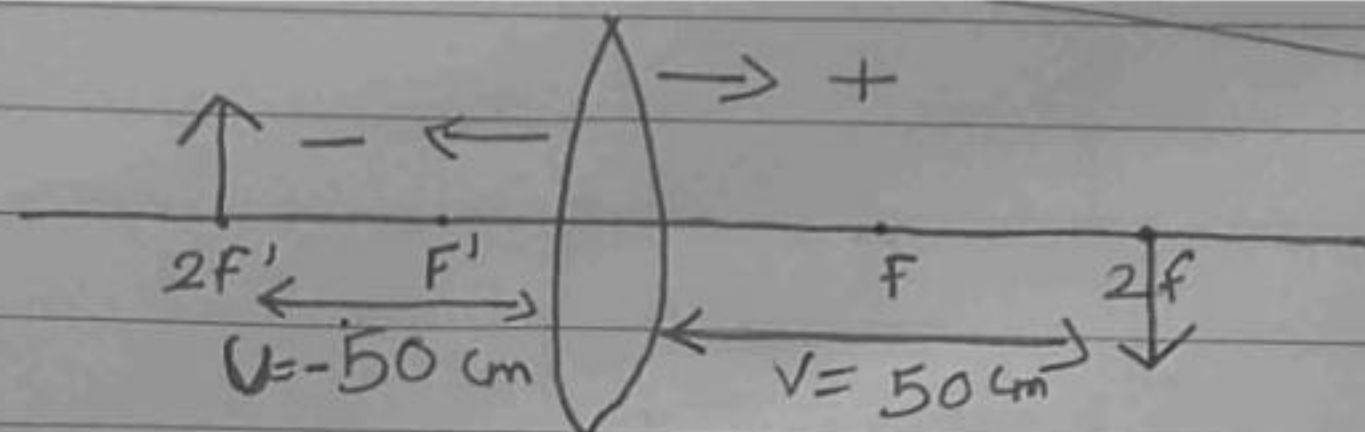
$$f = \frac{10}{3} = 3.33$$

$\therefore$ It is ³convex lens because f is +ve

Ans 4

$$U = -50 \text{ cm}$$

$$V = 50 \text{ cm}$$



$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} - \left(\frac{1}{-50} \right)$$

$$\frac{1}{f} = \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25}$$

$$f = 25 \text{ cm}$$

$$f = \frac{25}{100} \text{ m}$$

$$f = \text{0.25 m}$$

Ans. 4

$$\text{Power} = \frac{1}{\text{Focal length}}$$

$$P = \frac{1}{f}$$

$$P = \frac{1}{0.25} = +4D$$